

HW 4.

$$1.) H_0: y_i = \beta_0 + \varepsilon_i$$

$$H_1: y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_{p-1} x_{(p-1)i} + \varepsilon_i$$

$$F = \frac{MSR}{MSE} \Leftrightarrow F = \frac{(SSE_{H_0} - SSE_{H_1}) / (df_{SSE_{H_0}} - df_{SSE_{H_1}})}{\frac{SSE_{H_1}}{df_{SSE_{H_1}}}}$$

$$SSE_{H_0} = \sum (e_i)^2 = \sum (y_i - \hat{y}_i)^2 = SST$$

$$\hat{y}_i = \hat{\beta}_0 = \bar{y}$$


$$\begin{aligned} SSE_{H_1} &= \sum e_i^2 = \sum (y_i - \hat{y}_i)^2 \\ &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_{p-1} x_{(p-1)i})^2 \\ &= SST - SSR_F \end{aligned}$$

$$\begin{aligned} \frac{SSE_{H_0} - SSE_{H_1}}{(n-1) - (n-p)} &= \frac{SST - (SST - SSR)}{p-1} \\ &= \frac{SSR}{p-1} = MSR \checkmark \end{aligned}$$

Den: $\frac{SSE_{H_1}}{n-p} \geq \frac{SSE_F}{n-p} = MSE \checkmark$

2.) 5, we remove one because it is the baseline.

3.) Not necessarily, because one category being insignificant does not mean the entire predictor is. We should look at the global F-test.